THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The nth term of the pattern $4, 11, 18, 25, \ldots$ is $a + (n-1)d$. The 12th term is: 1

(a) 82 **(b)** 88 **(c)** 81 **(d)** 74

2. What is the smaller angle between the hands of a clock at 3:30? 1

(a) 75 **(b)** 85 **(c)** 70 **(d)** 80

3. The smallest 4-digit number with digits 3, 1, 4, 2 (each used once) that is EVEN is: 1

(a) 1225 **(b)** 1243 **(c)** 1324 **(d)** 1234

4. The total of the two largest values in the table is: 1

Item	Number
Apple	35
Banana	70
Orange	65
Grapes	65
Mango	60

(a) 130 **(b)** 140 **(c)** 135 **(d)** 145

5. The number of factors of 48 is: 1

(a) 11 **(b)** 9 **(c)** 10 **(d)** 12

6. In the pattern $3, 8, 13, 18, \ldots$ which term number equals 88? 1

(a) 19 **(b)** 20 **(c)** 18 **(d)** 17

7. The minute hand starts at 12 and stops at 4. The angle it turns through (clockwise) is: 1

(a) 90 **(b)** 126 **(c)** 150 **(d)** 120

8. The number of 3-digit palindromes is: 1

(a) 100 **(b)** 99 **(c)** 90 **(d)** 80

9. A pictograph has one symbol = 15. The number of symbols for A is 9 and for B is 4. How many more does A have than B? 1

(a) 60 **(b)** 76 **(c)** 75 **(d)** 90

10. The HCF of 36 and 48 is: 1

(a) 11 **(b)** 12 **(c)** 14 **(d)** 13

11. What is the sum of the first 12 even numbers $2 + 4 + 6 + \ldots$? 1

(a) 144 **(b)** 157 **(c)** 168 **(d)** 156

12. The hour hand of a clock turns through 150°. How many hours have passed? 1

(a) 7 **(b)** 5 **(c)** 4 **(d)** 6

13. The greatest 5-digit palindrome with first digit 2 is: 1

(a) 29982 **(b)** 30002 **(c)** 30092 **(d)** 29992

14. A bar of height 6 cm in a graph with scale 1 cm = 15 units, and another bar of 5 cm. Together they show: 1

(a) 180 **(b)** 165 **(c)** 150 **(d)** 175

15. The LCM of 12 and 18 is: 1

(a) 48 **(b)** 54 **(c)** 24 **(d)** 36

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the nth term formula of $5, 8, 11, 14, \ldots$ and the 20th term. 2

17. Find the smaller angle between the clock hands at 5:30. 2

18. Find the greatest and the smallest 5-digit numbers formed from 4, 9, 1, 7, 3, each digit once. 2

19. Find the difference between the highest and lowest and the total of all. 2

Item	Number
Apple	25
Mango	70
Orange	70
Guava	55
Grapes	70

20. Find the HCF and LCM of 24 and 36 using prime factorisation. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. How many triangles' worth of sticks: a row of 12 triangles needs 3 sticks for the first triangle and 2 more for each new one. Find the sticks needed and the number of triangles that can be made with 35 sticks. 3

22. At what angle are the hands of a clock inclined at 1:30? Is the angle acute, obtuse or reflex (looking at the larger angle too)? 3

23. Perform two Kaprekar steps on 1795. 3

24. Find the fraction of students liking Kabaddi and the number who do not like it. 3

Sport	Students
Kabaddi	40
Chess	65
Tennis	45
Cricket	15

OR

Find the prime factorisation of 180 and the number of its factors. 3

25. Find the sum of all odd numbers from 1 to 59. Hence find the sum of all even numbers from 2 to 60. 3

OR

A ray OA is turned about O: first 60° anticlockwise, then 70° anticlockwise, then 40° clockwise. Find the net angle of turn from the original position. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A pattern of squares: the first figure has 5 squares and each next figure has 4 more. Find the number of squares in the 10th figure, and the first figure that has more than 91 squares. 5

OR

A clock shows 3:00. Through how many degrees must the minute hand turn to make the time 3:40? Through how many degrees has the hour hand turned? Find the angle between the hands then. 5

27. A palindrome of the form 2__2 has its middle two digits equal and the digit sum 10. Find the number and check that it reads the same backwards. Add 11 to it and see if the answer is a palindrome. 5

OR

The pictograph (one symbol = 12) shows 3 symbols on Monday, 6 on Tuesday and 9 on Wednesday. On Thursday the number is the average of these three days. Find the number of items on each day and on Thursday. 5

END OF QUESTION PAPER **Total: 50 Marks**

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